

CORE 1A

Motion in 2D - Projectile Motion

Detailed declarative teaching: R1-R3 first, then five labelled extensions.

How to use this Core

For each concept: read the inferential jump, follow the construction, study the worked anchor, inspect the wrong path, then attempt the exit task before reading the model answer.

1. R1 - Independent x/y components, one shared clock

Inferential jump

Perpendicular component equations can be solved separately, but they describe one object at one physical time.

Stepwise construction

1. Choose one fixed x-y frame, origin, sign convention, and $t=0$ for the entire event.
2. Represent position, velocity, and acceleration as signed x and y components.
3. Keep x quantities in x equations and y quantities in y equations, while using one common elapsed time.
4. Recombine only component values that refer to the same frame and the same instant.

Worked conceptual anchor

Anchor

At $t=3$ s a particle has $v_x=6$ m/s and $v_y=-8$ m/s. These are one velocity state only because both components use the same frame and the same $t=3$ s.

Plausible wrong path -> repair

Wrong: independent components may use different elapsed times. Repair: independence separates axis equations; it does not give the same object two clocks.

Exit task

A particle has $v_x=6$ m/s and $v_y=-8$ m/s at $t=3$ s. Explain what 'independent components' means and what must remain shared.

Work before reading the model answer

Model answer

Solve the axes separately, but keep one coordinate frame and one common physical time. Recombine only simultaneous components.

2. R2 - Constant-acceleration kinematics component by component

Inferential jump

The one-dimensional constant-acceleration equations apply independently to x and y only when each acceleration component is constant over the interval.

Stepwise construction

1. Check separately whether a_x and a_y are constant over the interval.
2. Use $v_x = u_x + a_x t$ and $v_y = u_y + a_y t$ with the same t .
3. Use $\Delta x = u_x t + 0.5 a_x t^2$ and $\Delta y = u_y t + 0.5 a_y t^2$ with the same t .
4. If one axis determines an event time, reuse that event time on the other axis.

Worked conceptual anchor

Anchor

For $u = (4, 6)$ m/s and $a = (0, -2)$ m/s², after 2 s the velocity is $v = (4, 2)$ m/s.

Plausible wrong path -> repair

Wrong: insert the magnitude of acceleration into both axis equations. Repair: use only the signed component belonging to that axis.

Exit task

A particle starts with $u = (0, 5)$ m/s and constant $a = (10, 4)$ m/s². Set up x and y displacement equations at common time t .

Work before reading the model answer

Model answer

$\Delta x = 0.5(10)t^2$ and $\Delta y = 5t + 0.5(4)t^2$, with the same t .

3. R3 - Select the gravity-only projectile model before using projectile consequences

Inferential jump

Ideal projectile motion is ordinary 2D component kinematics with $a_x=0$ and $a_y=-g$ (for +y upward), selected only after retained forces are checked.

Stepwise construction

1. Decide whether the object is in ideal near-Earth free flight after release; neglect air resistance only when allowed.
2. Set $a_x=0$ and $a_y=-g$. Use $v_x=u_x$, $v_y=u_y-gt$, $\Delta x=u_x t$, $\Delta y=u_y t-0.5gt^2$.
3. Translate the requested event into a condition: apex $v_y=0$; same-height return $\Delta y=0$; unequal-height landing uses actual Δy .
4. At the event time, reconstruct speed, direction, position, or range from simultaneous components.

Worked conceptual anchor

Anchor

For a horizontal launch from a cliff: $a_x=0$, $a_y=-g$; $v_x=u$; $v_y=-gt$; $\Delta x=ut$; $\Delta y=-0.5gt^2$.

Plausible wrong path -> repair

Wrong: at the apex the whole velocity and acceleration are zero. Repair: only $v_y=0$; v_x remains nonzero and $a_y=-g$.

Exit task

A stone is launched horizontally from a cliff with speed u . Write its component acceleration, velocity, and displacement relations.

Work before reading the model answer

Model answer

$a_x=0$, $a_y=-g$; $v_x=u$, $v_y=-gt$; $\Delta x=ut$, $\Delta y=-0.5gt^2$; one common t describes the same stone.

4. E1 - EXTENSION: instantaneous velocity direction from $x(t)$, $y(t)$

Inferential jump

Direction of motion comes from tangent velocity (dx/dt , dy/dt), not the position vector (x,y) .

Stepwise construction

1. Identify that the target is instantaneous velocity direction.
2. Differentiate $x(t)$ and $y(t)$ separately to obtain $v_x(t)$ and $v_y(t)$.
3. Evaluate both at the same requested time.
4. Use the signed ratio v_y/v_x and check the quadrant; confirm the velocity is tangent to the path.

Worked conceptual anchor

Anchor

If $x=4t$ and $y=10t-t^2$, then at $t=2$ s, $v=(4,6)$ m/s.

Plausible wrong path -> repair

Wrong: use y/x for direction of motion. Repair: y/x is a position-ray direction; use derivatives for the tangent velocity.

Exit task

If $x=5t$ and $y=12t-t^2$, state the operations needed to find velocity direction at $t=2$ s.

Work before reading the model answer

Model answer

Differentiate: $v_x=5$, $v_y=12-2t$; evaluate at $t=2$ s, then use the signed velocity-component ratio and quadrant.

5. E2 - EXTENSION: projectile rebound after an energy-changing impact

Inferential jump

A kinetic-energy retention fraction multiplies v^2 . After impact creates a new launch state, the new vertical component controls the next rise height.

Stepwise construction

1. Treat impact as an event boundary: pre-impact flight ends and a new launch state begins.
2. For unchanged mass, convert $K_{\text{after}} = f K_{\text{before}}$ into $v_{\text{after}}^2 = f v_{\text{before}}^2$.
3. Resolve the post-bounce velocity using the stated direction.
4. Use $H = (v_y^2)/(2g)$ for the next gravity-only rise and check that energy loss cannot increase the height.

Worked conceptual anchor

Anchor

If 49% of kinetic energy remains, 70% of the speed remains because $v_{\text{after}}/v_{\text{before}} = \sqrt{0.49} = 0.70$.

Plausible wrong path -> repair

Wrong: half the kinetic energy means half the speed. Repair: kinetic energy is proportional to speed squared.

Exit task

A ball keeps 64% of its kinetic energy and rebounds at angle α . State the chain needed to find its new maximum height.

Work before reading the model answer

Model answer

Use $v_{\text{after}}^2 = 0.64 v_{\text{before}}^2$, then $v_y = v_{\text{after}} \sin(\alpha)$, then $H = v_y^2/(2g)$.

6. E3 - EXTENSION: projectile motion when gravity changes after the apex

Inferential jump

A change in constant gravity creates two stages joined by one boundary state. Position and velocity are inherited; acceleration changes.

Stepwise construction

1. Split the trajectory at the named event boundary (here, the apex).
2. Use the ordinary projectile model only up to the boundary to obtain the apex state.
3. Start stage 2 from the apex with $v_y=0$ and the new downward acceleration g_2 .
4. Compute the new descent time and convert it to horizontal distance using unchanged v_x .

Worked conceptual anchor

Anchor

From the same apex height, if gravity doubles after the apex, descent time becomes $1/\sqrt{2}$ of its ordinary value.

Plausible wrong path -> repair

Wrong: once gravity changes, recompute the whole trajectory with the new gravity. Repair: preserve the stage-1 boundary state and apply g_2 only after the event.

Exit task

If post-apex gravity is four times larger, how does descent time from the same apex height compare with the ordinary descent?

Work before reading the model answer

Model answer

It is half as long because fall time from rest through the same height scales as $1/\sqrt{g}$.

7. E4 - EXTENSION: trajectory equation through a specified point

Inferential jump

Eliminating the shared time gives a path equation $y(x)$, but point membership and branch/apex location are separate decisions.

Stepwise construction

1. License the gravity-only projectile model first.
2. Eliminate t from $x = u \cos(\theta)t$ and $y = u \sin(\theta)t - 0.5gt^2$.
3. Use $y = x \tan(\theta) - g x^2 / [2u^2 \cos^2(\theta)]$ and impose the specified point.
4. Use v_y or local slope dy/dx at the point to decide whether the projectile is rising or falling.

Worked conceptual anchor

Anchor

For a known point on a 45-degree trajectory, a negative local slope or negative v_y shows that the point is after the apex.

Plausible wrong path -> repair

Wrong: if a point satisfies the trajectory equation, the apex must be after it. Repair: point membership alone does not determine the branch.

Exit task

What two checks are needed when asked whether a projectile passes through P and has already crossed its apex there?

Work before reading the model answer

Model answer

First impose P in the trajectory equation. Then use v_y or local slope at P to decide whether the path is descending.

8. E5 - EXTENSION: projectile motion with linear drag

Inferential jump

Component splitting survives, but the constant-acceleration law fails where acceleration depends on velocity.

Stepwise construction

1. Identify the force model. For horizontal linear drag, $m \, dv_x/dt = -c \, v_x$.
2. Reject constant horizontal velocity and solve $v_x(t) = v_{x0} \exp[-(c/m)t]$.
3. Integrate velocity to get $x(t) = v_{x0}(m/c)[1 - \exp(-ct/m)]$.
4. Check the zero-drag/short-time limit, where x approaches $v_{x0} t$.

Worked conceptual anchor

Anchor

If $m/c = 2 \, \text{s}$, $v_{x0} = 100 \, \text{m/s}$ and $t = 2 \, \text{s}$, $v_x = 100/e$ and $x = 200(1 - 1/e)$, both below the no-drag values.

Plausible wrong path -> repair

Wrong: projectile horizontal velocity is always constant. Repair: that is true only when the horizontal force is zero.

Exit task

If a medium exerts $F_x = -c \, v_x$, why is $x = v_{x0} t$ not generally valid?

Work before reading the model answer

Model answer

Because $a_x = -(c/m)v_x$ is not zero. Solve exponential velocity decay first, then integrate to obtain displacement.